

Stock Trading AI Analyst

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April 2026

The Problem

- New traders enter the market **without practical experience**
- Most lose money due to poor risk understanding
- Existing tools are too advanced **or** too simplified
- No safe environment to learn before risking real money

Project Goals

- Simulate real trading with fake money
- Provide AI-generated buy, sell, or hold recommendations
- Display charts, portfolio data, and gain/loss feedback
- Offer practice and simulation modes for guided learning
- Store all user activity persistently

System Architecture

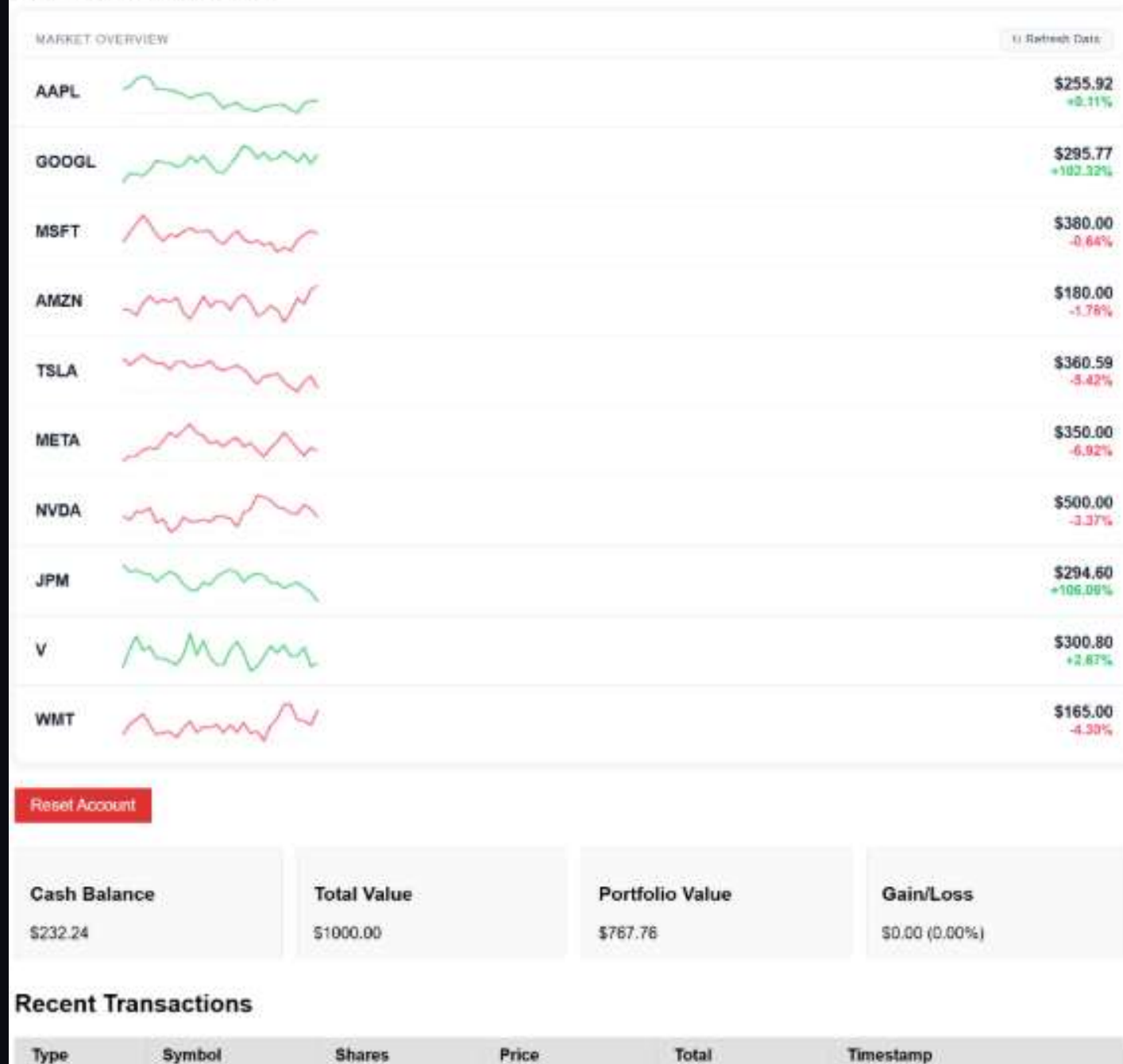
- **Flask web app** - routes, sessions, multi-page UI
- **SQLite database** - users, accounts, holdings, transactions
- **Bokeh** - interactive stock charts
- **Ollama** - LLM based predictions + fallback

Core Feature: Trading

- Register and log in with a personal account
- Start with \$1,000 in simulated funds
- Search stocks by symbol
- Buy and sell shares through the web interface
- Portfolio and balance update immediately after each trade

Dashboard

Account overview and recent activity



Core Feature: AI Recommendations

- Powered by Llama model 3.2 running locally
- Returns BUY / SELL / HOLD with confidence and risk level
- Includes a short rationale for the recommendation
- Available on every stock detail page

AAPL - AAPL Corporation

About: Apple develops consumer electronics, software, and services. Known for iPhones, Macs, and the App Store ecosystem.

Price: \$255.92 | Volume: 31289369

Quick Trade

Action: Number of Shares:

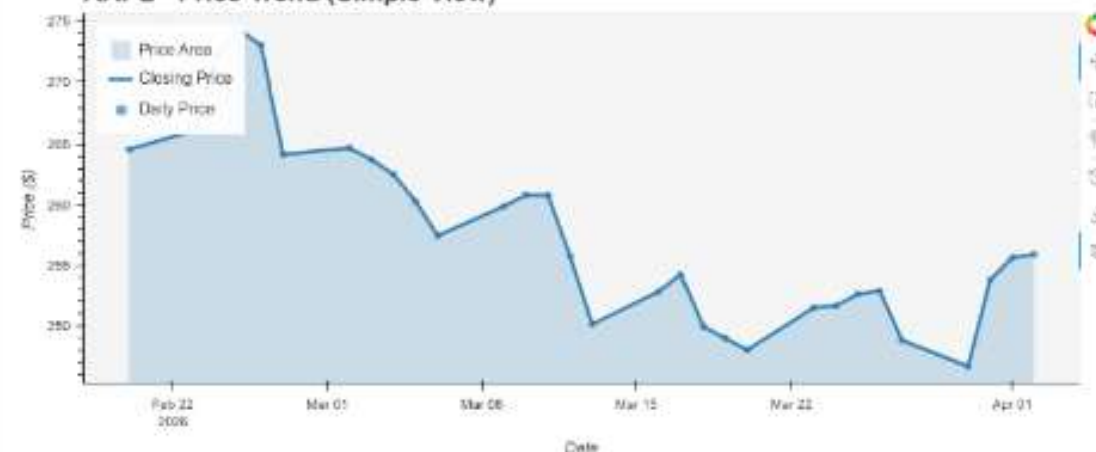
AI Recommendation

Action: HOLD
Confidence: 62.0%
Risk: medium
Expected Change: -0.01%

AAPL's recent price drop and tight 30-day range suggest caution, while its trend strength is relatively weak (-0.0327). A moderate risk level suggests a potential short-term correction.

Interactive Charts

AAPL - Price Trend (Simple View)



Easy-to-understand price trend. Shows all price data on hover.

Recent Candles (last 10 days)

Date	Open	High	Low	Close	Volume
2026-04-02	\$254.20	\$256.13	\$250.85	\$255.92	31289369
2026-04-01	\$254.08	\$256.18	\$253.33	\$255.83	40059432
2026-03-31	\$247.91	\$255.48	\$247.10	\$253.79	49598091

Practice and Simulation Modes

- Practice Mode - synthetic companies, news events, difficulty levels
- Simulation Mode - replay real historical market data over time
- Both modes are isolated from the main trading account
- Designed for learning without affecting real portfolio state

Practice Mode

Unlimited practice with randomly generated stocks and market scenarios.

Practice Session Active - Difficulty: Hard

Day 22 of 90
Date: 2026-01-27
Progress: 24.4%

Today's News

- GEJH announces quarterly results
- DVXM stock upgraded by analysts
- STZA secures major partnership deal
- ESRK expands into new markets

Advance 1 Day

Advance 1 Week

Advance 1 Month

End Practice

Current Portfolio

Cash Balance	Total Value	Portfolio Value	Gain/Loss
\$67.96	\$846.48	\$778.52	\$-153.52 (-15.35%)

Holdings

Symbol	Shares	Purchase Price	Current Price	Value	Gain/Loss
GEJH	6	\$84.10	\$86.74	\$400.44	\$-104.16
UVBU	2	\$213.72	\$189.04	\$378.08	\$-49.36

Available Stocks (20 companies)

These companies are randomly generated for practice - each session is unique!

GEJH	MDHS	SGYJ	DVXM	UVBU	STZA	SNCC	SLQL	QCPW	NCLT	SHLG	ACYZ
SDMW	ESRK	AHRR	SHFJ	UGCM	QCCG	ANVY	GDPZ				

Challenges Overcome

- API rate limits - caching and fallback behavior added
- AI availability - rule-based fallback keeps app running
- Full-stack integration - connected data, AI, trading, and UI into one system
- Scope management - prioritized core flows over unfinished features

Future Enhancements

- Complete multi-account workflow for the main UI
- Improve AI performance tracking and reporting
- Add expanded educational tools (alerts, quizzes, learning modules)
- Stronger account security and portfolio analytics
- Load times for pages / preloading pages & AI recommendations.

Summary

- Functional educational trading platform in Python + Flask
- Integrated real market data, AI analysis, and persistent storage
- Covered practice and simulation modes for beginner learning

Questions?

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Source: github.com/MichaelLudwikowski/MLud-CSU-Senior-Project